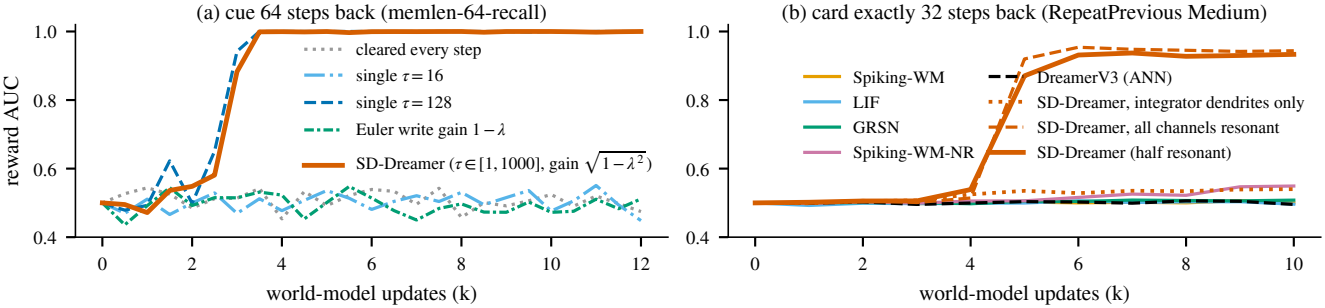
(a) cue 64 steps back (memlen-64-recall)
reward AUC
1.0
0.8
0.6
0.4
0
2
4
6
8
10
12
world-model updates (k)
cleared every step
single $\tau = 16$
single $\tau = 128$
Euler write gain $1 - \lambda$
SD-Dreamer ($\tau \in [1, 1000]$, gain $\sqrt{1-\lambda^2}$)
(b) card exactly 32 steps back (RepeatPrevious Medium)
1.0
0.8
0.6
0.4
0
2
4
6
8
10
world-model updates (k)
Spiking-WM
LIF
GRSN
Spiking-WM-NR
DreamerV3 (ANN)
SD-Dreamer, integrator dendrites only
SD-Dreamer, all channels resonant
SD-Dreamer (half resonant)